

AI Usage Reflection

PMP Solutions

Throughout the development of the PMP Solutions AI Fraud Intelligence Platform, our team used Artificial Intelligence (AI) as a collaborative development assistant to improve our understanding, productivity, problem-solving, and the overall quality of our solution. Rather than replacing our own work, AI supported us by accelerating research, explaining technical concepts, debugging code, improving documentation, and helping us make informed technical decisions throughout the Software Development Life Cycle (SDLC).

During planning, AI helped us refine the project scope, understand the fraud detection problem, compare technologies such as React, Plotly, and Streamlit, and select the most suitable architecture.

AI helped us understand the PaySim dataset by explaining transaction types, transaction amounts, sender and receiver balances, fraud labels, account identifiers, and how these variables relate to fraudulent behaviour. It also guided us through data cleaning by checking missing values, duplicate records, incorrect data types, zero-value transactions, negative transaction amounts, and explaining why these steps improve model quality.

For exploratory data analysis, AI recommended meaningful visualisations, including fraud versus legitimate transactions, transaction type distributions, fraud cases by transaction type, transaction amount distributions, and feature importance charts. AI also helped us interpret the results and explain their business significance.

During feature engineering, AI assisted us in creating Balance Change, Balance Change Ratio, High Value Transaction, Account Emptied, New Destination Account, Risk Score, and other predictive features while explaining why certain features should be retained or removed to prevent data leakage.

AI supported our work with Logistic Regression, Decision Trees, Random Forest, and XGBoost by explaining algorithms, assisting with hyperparameter tuning, handling class imbalance, interpreting Accuracy, Precision, Recall, F1-Score, ROC-AUC, confusion matrices, and integrating SHAP explainability.

Throughout development, AI helped us debug Python code, resolve Google Colab and Jupyter Notebook issues, fix package installation problems, correct file paths, troubleshoot Plotly and Streamlit code, structure our dashboard, organise our GitHub repository, prepare our README, SDLC, executive summary, presentation, and project documentation.

Most importantly, AI acted as a learning and development partner rather than replacing our work. Every recommendation, explanation, and code suggestion was reviewed, tested,

validated, and adapted by our team before being incorporated into the final solution. Our team remained responsible for implementation, testing, verification, and presentation.

Overall, AI enhanced our technical understanding, increased development efficiency, improved code quality, strengthened our documentation, and enabled us to deliver a more accurate, explainable, and professional AI-powered fraud detection platform.